

Random-Key Optimizer and Linearization for the Quadratic Multiple Constraints Variable-Sized Bin Packing Problem

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Additional Computational Results on Benchmark Instances

This supplementary material expands upon the computational experiments presented in the main article by providing performance evaluations on standard literature benchmark instances for the QMC-VSBPP. This material complements the results reported in the main body of the published article, which were executed on a specific set of symmetric instances available in our repository¹.

To enable standard comparisons, the mathematical models (original and linearized) and the proposed Random-Key Optimizer (RKO) algorithm were evaluated on benchmark instances from the literature (asymmetric instances). These results validate the exact approach and the metaheuristic's performance under the standard testing environment recognized in the literature.

Table 1 presents the upper (UB) and lower (LB) bounds and optimality gaps for the original and linearized models using the Gurobi solver. Additionally, it reports the best-known solutions (BKS) from the literature, along with the best and average solution values obtained by the RKO, as well as

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¹<https://github.com/nataliaalves03/RKO-AC0>

the optimality gap of the best RKO solution with respect to the best lower bound provided by the mathematical models.

Table 1: Computational results of the original and linearized models, and the RKO algorithm, evaluated on asymmetric benchmark instances.

Instance			Original model			Linearized model			RKO		
ID	n	BKS	LB	UB	Gap(%)	LB	UB	Gap(%)	Best	Avg	Gap(%)
1	25	7746	6830	7746	11.8	7516	7746	3.0	7746	7746	3.0
2	25	7581	6520	7625	14.5	7294	7600	4.0	7581	7581	3.8
3	25	7465	6514	7584	14.1	7437	7465	0.4	7465	7465	0.4
4	25	12009	9640	12009	19.7	10368	11973	13.4	11933	11938	13.1
5	25	11561	9492	11988	20.8	9530	12061	21.0	11543	11571	17.4
6	25	11558	9377	11871	21.0	9577	11876	19.4	11430	11437	16.2
7	25	5666	5436	5666	4.1	5666	5666	0.0	5666	5666	0.0
8	25	5367	5108	5367	4.8	5367	5367	0.0	5367	5367	0.0
9	25	5367	5033	5394	6.7	5367	5367	0.0	5367	5367	0.0
10	25	8302	6676	8381	20.3	7223	8302	13.0	8302	8302	13.0
11	25	7882	6453	8245	21.7	6892	8266	16.6	7864	7864	12.4
12	25	7718	6220	7821	20.5	6748	8128	17.0	7697	7697	12.3
13	25	11410	9318	11346	17.9	9865	11423	13.6	11325	11325	12.9
14	25	10919	8998	11063	18.7	9700	10919	11.2	10919	10919	11.2
15	25	10853	8926	10943	18.4	9582	10908	12.2	10768	10768	11.0
16	25	18430	15645	18479	15.3	16422	18703	12.2	18015	18138	8.8
17	25	18252	14786	18792	21.3	15116	18595	18.7	18015	18105	16.1
18	25	17194	13704	17680	22.5	14140	17706	20.1	17217	17217	17.9
19	25	8516	7278	8635	15.7	7904	8646	8.6	8516	8516	7.2
20	25	8206	7429	8222	9.6	7646	8239	7.2	8206	8206	6.8
21	25	8144	7381	8164	9.6	7654	8206	6.7	8144	8144	6.0
22	25	11842	9508	11865	19.9	10180	11919	14.6	11775	11775	13.5
23	25	11476	9173	11572	20.7	9925	11674	15.0	11446	11464	13.3
24	25	11144	9238	11526	19.9	9729	11144	12.7	11144	11144	12.7
25	50	26229	10131	26710	62.1	10614	26860	60.5	26161	26170	59.4
26	50	24852	10262	25387	59.6	9558	25290	62.2	24767	24769	58.6
27	50	25101	9964	25762	61.3	9447	25351	62.7	24579	24672	59.5
28	50	33871	15433	34557	55.3	14312	34177	58.1	33796	33948	54.3
29	50	33380	15262	35099	56.5	14584	35027	58.4	33404	33699	54.3
30	50	32929	15006	34570	56.6	16199	34713	53.3	32796	32958	50.6
31	50	22316	6970	22992	69.7	7002	22847	69.4	22264	22268	68.6
32	50	20656	5724	20865	72.6	5153	21622	76.2	20376	20388	71.9
33	50	20445	6110	21124	71.1	5695	21052	72.9	20376	20377	70.0
34	50	27104	9567	27684	65.4	9249	27799	66.7	27019	27019	64.6
35	50	26514	9757	27760	64.9	8973	27883	67.8	26302	26588	62.9
36	50	25942	9212	27312	66.3	8859	27262	67.5	25701	25703	64.2
37	50	32921	15750	33367	52.8	15207	33277	54.3	32778	32821	51.9

Instance			Original model			Linearized model			RKO		
ID	n	BKS	LB	UB	Gap(%)	LB	UB	Gap(%)	Best	Avg	Gap(%)
38	50	32812	15428	33341	53.7	15131	33014	54.2	32437	32495	52.4
39	50	32600	15219	33521	54.6	12327	32974	62.6	32135	32246	52.6
40	50	45919	25296	47880	47.2	24140	46759	48.4	45939	46026	44.9
41	50	46056	24424	48655	49.8	24530	47361	48.2	45793	45927	46.4
42	50	44960	22410	46965	52.3	22508	47272	52.4	44356	44506	49.3
43	50	27588	10818	28067	61.5	11494	28591	59.8	27418	27490	58.1
44	50	26792	11037	27606	60.0	9514	27202	65.0	26373	26380	58.2
45	50	26902	11092	26748	58.5	9235	27127	66.0	26225	26305	57.7
46	50	33543	15015	34633	56.6	15101	34910	56.7	33211	33310	54.5
47	50	33529	15125	34279	55.9	14668	34205	57.1	33152	33270	54.4
48	50	33134	16552	34843	52.5	15534	34029	54.4	32498	32498	49.1
49	100	100694	17569	110411	84.1	17362	108590	84.0	100933	101104	82.6
50	100	97933	17295	116181	85.1	17333	105474	83.6	97754	98009	82.3
51	100	97922	17312	108374	84.0	17312	107425	83.9	97459	97663	82.2
52	100	116348	27410	130243	79.0	27811	133263	79.1	117251	117550	76.3
53	100	116494	27421	137478	80.1	27392	131824	79.2	116851	117085	76.5
54	100	115508	27374	132896	79.4	27375	129415	78.8	115165	115542	76.2
55	100	92689	10349	105148	90.2	10418	100844	89.7	92112	92163	88.7
56	100	88587	9480	98029	90.3	9391	96181	90.2	88413	88556	89.3
57	100	88738	9340	98524	90.5	9340	97268	90.4	88369	88566	89.4
58	100	102688	16344	113344	85.6	16016	113562	85.9	102232	102445	84.0
59	100	101898	15805	118285	86.6	15749	116448	86.5	101353	101696	84.4
60	100	99941	15410	115153	86.6	15287	110176	86.1	99942	100136	84.6
61	100	115685	26586	122822	78.4	26428	119764	77.9	114782	114978	76.8
62	100	115487	25579	121367	78.9	26357	121290	78.3	114448	114625	77.0
63	100	114982	25026	120935	79.3	25699	120589	78.7	113651	113897	77.4
64	100	141233	45842	150622	69.6	46027	157607	70.8	141835	142429	67.5
65	100	141348	45732	148827	69.3	47136	156204	69.8	141985	142451	66.8
66	100	140308	43802	150855	71.0	43870	150656	70.9	139699	140116	68.6
67	100	103445	18183	111642	83.7	18286	111245	83.6	102776	103114	82.2
68	100	102140	18185	110188	83.5	18272	107806	83.1	101464	101586	82.0
69	100	101713	17907	110608	83.8	17939	110522	83.8	100520	100852	82.2
70	100	115442	27005	129520	79.1	26992	128081	78.9	116203	116281	76.8
71	100	115037	26740	125292	78.7	26667	125080	78.7	115546	115621	76.9
72	100	114366	26516	125246	78.8	26530	122596	78.4	114676	114956	76.9
73	200	392121	30470	410914	92.6	30470	411350	92.6	393587	394009	92.3
74	200	386144	30470	397576	92.3	30470	410449	92.6	387061	387451	92.1
75	200	387032	30470	399244	92.4	30470	411355	92.6	386260	386579	92.1
76	200	422785	50565	437750	88.4	50565	451844	88.8	425838	426799	88.1
77	200	423046	50565	451856	88.8	50565	456017	88.9	426793	427034	88.2
78	200	421503	50565	465082	89.1	50565	458103	89.0	422572	423529	88.0
79	200	375838	16107	391804	95.9	16107	397684	95.9	375100	375382	95.7
80	200	367479	14459	374381	96.1	14460	396933	96.4	366750	367556	96.1

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81	200	367545	14442	380953	96.2	14442	399015	96.4	366778	367187	96.1
82	200	395569	27553	421473	93.5	27553	421986	93.5	395616	396155	93.0
83	200	394290	26310	418105	93.7	26310	425222	93.8	393757	394537	93.3
84	200	390476	25987	417815	93.8	25987	429708	94.0	390814	391098	93.4
85	200	422355	45272	429213	89.5	45272	433427	89.6	421662	422093	89.3
86	200	422130	45229	431868	89.5	45229	434254	89.6	421082	421597	89.3
87	200	421546	44800	432021	89.6	44800	434064	89.7	418649	418994	89.3
88	200	472193	85027	496293	82.9	85028	502579	83.1	476494	477158	82.2
89	200	472472	85027	495268	82.8	85028	507779	83.3	476701	477587	82.2
90	200	470972	79785	502535	84.1	79786	501049	84.1	473152	473454	83.1
91	200	398077	31556	412618	92.4	31557	413792	92.4	397681	398155	92.1
92	200	395183	31556	413040	92.4	31557	414904	92.4	394758	395347	92.0
93	200	394632	31490	411384	92.3	31491	417731	92.5	392006	392641	92.0
94	200	420590	49357	446008	88.9	49358	449800	89.0	424533	425406	88.4
95	200	420896	49196	443255	88.9	49196	444650	88.9	423159	424166	88.4
96	200	420189	48657	447362	89.1	48658	446515	89.1	421821	422442	88.5

An analysis of these exact results reveals that the original model obtained the best objective values for instances 13 and 45. Meanwhile, the linearized model achieved the best objective value for instance 4 and successfully proved optimality for instances 7, 8, and 9.

Overall, the RKO algorithm achieved solutions equal to or strictly better than the BKS in 73% of the evaluated instances, notably maintaining this strong performance across the largest instance sets ($n = 200$). Furthermore, in the minority of instances where RKO was outperformed, the relative difference between the RKO’s best solution and the BKS remained consistently below 1%.

Data Availability

To allow for exact reproducibility of the published results, the asymmetric benchmark instances and the symmetric dataset have been made publicly available in our repository at: <https://github.com/nataliaalves03/RKO-ACO>.